

# **SOCIOL 723: Social Statistics II**

Week 13, Causal Machine Learning

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# Today

Why Naive ML Fails

Double Machine Learning

Heterogeneous Treatment Effects

★ = essential, you must understand this

★ = for understanding, not examined

## Putting the Two Halves Together ★

- **Weeks 5–11** gave us identification: what must be true for a parameter to be causal. Almost every design ended with a nuisance function to estimate, a propensity score, an outcome regression, a first stage, a conditional trend.
- **Week 12** gave us flexible estimators for exactly such functions, which work with many covariates and unknown functional form.
- **The obvious move, plug ML into the causal estimator, fails.** Today is about why, and what fixes it.

### The payoff

We get to relax the functional-form assumptions that identification arguments have been quietly leaning on all semester, AJR's linear geography control, the linear propensity score, the additive conditional trend, *without* giving up  $\sqrt{n}$  inference.

# The Problem

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## The Partially Linear Model ★

To see the problem without letting notation hide it, start with the simplest causal case there is: *one* treatment effect to estimate, but completely flexible relationships between the controls, the treatment, and the outcome. That is the **partially linear model**, the workhorse for the rest of the course:

$$\begin{aligned} Y_i &= \theta D_i + g(W_i) + \epsilon_i, & E[\epsilon_i \mid D_i, W_i] &= 0 \\ D_i &= m(W_i) + v_i, & E[v_i \mid W_i] &= 0. \end{aligned}$$

- $\theta$  is the target: the effect of  $D$  holding  $W$  fixed, and it is assumed constant (relaxed in the last section).
- $g$  and  $m$  are **nuisance functions**; we do not care about them, but we must handle them. They may be highly nonlinear, and  $W$  may be high-dimensional.
- Under conditional ignorability given  $W$  (Week 5),  $\theta$  is the ATE.
- Week 1 solved this when  $g$  and  $m$  are linear and low-dimensional. The question now is what happens when we estimate them with lasso or a forest.

## Attempt 1: Regularization Bias ★

The naive approach: fit the partially linear model by ML to get  $\hat{g}$ , taking it from a joint fit of  $Y$  on  $(D_i, W_i)$  with  $D_i$  left *unpenalized* so that  $\hat{g}$  targets the structural  $g$ . Then regress  $Y_i - \hat{g}(W_i)$  on  $D_i$ .

$$\hat{\theta}_{\text{naive}} = \frac{\frac{1}{n} \sum_i D_i (Y_i - \hat{g}(W_i))}{\frac{1}{n} \sum_i D_i^2}.$$

Substituting the true model and rescaling,

$$\sqrt{n}(\hat{\theta}_{\text{naive}} - \theta) = \underbrace{\left(\frac{1}{n} \sum_i D_i^2\right)^{-1} \frac{1}{\sqrt{n}} \sum_i D_i \epsilon_i}_{\xrightarrow{d} \mathcal{N}(0, \Sigma)} + \underbrace{\left(\frac{1}{n} \sum_i D_i^2\right)^{-1} \frac{1}{\sqrt{n}} \sum_i D_i (g(W_i) - \hat{g}(W_i))}_{\text{regularization bias}}.$$

- Regularized estimators are *deliberately biased* (Week 12), and typically converge more slowly than the parametric  $n^{-1/2}$ .

## Attempt 1: Regularization Bias ★ (cont.)

- Then the second term does not vanish on the  $\sqrt{n}$  scale, the nuisance bias is of the same order as, or larger than, the sampling noise we are trying to describe.
- The point estimate may be fine; the inference is not. Confidence intervals are centred in the wrong place, and collecting more data does not fix it.
- **A trap worth naming.** If you instead regress  $Y$  on  $W$  alone you estimate  $\ell(W) = E[Y | W] = \theta m(W) + g(W)$ , which already contains the effect you are after. Subtracting that removes the signal. And even with the true  $\ell$ , regressing  $Y - \ell(W)$  on *raw*  $D_i$  does not recover  $\theta$ :  $D_i$  must be residualized too, which is exactly what the next slide does.

## Attempt 2: Partial Out Both Sides ★

Frisch–Waugh–Lovell says: residualize  $Y$  and  $D$  on  $W$ , then regress. With ML estimates  $\hat{g}$  and  $\hat{m}$ ,

$$\tilde{Y}_i = Y_i - \hat{\ell}(W_i), \quad \tilde{D}_i = D_i - \hat{m}(W_i), \quad \hat{\theta} = \frac{\frac{1}{n} \sum_i \tilde{D}_i \tilde{Y}_i}{\frac{1}{n} \sum_i \tilde{D}_i^2},$$

where  $\ell(W) = E[Y \mid W]$ .

The bias term now involves the **product** of the two estimation errors:

$$\text{bias} \approx \sqrt{n} \cdot \frac{1}{n} \sum_i (m(W_i) - \hat{m}(W_i)) (\ell(W_i) - \hat{\ell}(W_i)).$$

**This vanishes when**  $\|\hat{m} - m\|_2 \|\hat{\ell} - \ell\|_2 = o_p(n^{-1/2})$ , for which it is sufficient that *each* converge faster than  $n^{-1/4}$ . Lasso and forests can do that under sparsity or smoothness assumptions. **First-order bias has become second-order.**



# This Is Neyman Orthogonality Again ★

The residualized moment condition is

$$E[\psi(W_i; \theta, \eta)] = 0, \quad \psi = [(Y - \ell(W)) - \theta(D - m(W))] (D - m(W)),$$

and it satisfies

$$\left. \frac{\partial}{\partial \eta} E[\psi(W_i; \theta_0, \eta)] \right|_{\eta=\eta_0} = 0.$$

- **You met this property in Week 6:** the AIPW estimator's bias was the *product* of the two nuisance errors, so first-order mistakes in either one cancel. The partially linear case is the same statement for a different moment.
- And you derived it in **Week 1**: FWL is the linear special case, in which  $\hat{m}$  is an OLS projection and the “adaptivity” of  $\hat{\beta}_1$  to estimation error in the residuals is exactly orthogonality.
- **The thread of the whole course:** FWL  $\rightarrow$  AIPW  $\rightarrow$  DML. One idea, three levels of generality.

# DML

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## The Second Ingredient: Cross-Fitting ★

Orthogonality is not quite enough. If  $\hat{m}$  is fitted on the *same* observations where it is evaluated, its errors correlate with those units' outcomes, **overfitting bias**, which orthogonality does not remove.

### Cross-fitting

Split the sample into  $K$  folds. For each fold  $k$ : estimate  $\hat{\ell}^{(-k)}, \hat{m}^{(-k)}$  on the *other* folds, and form residuals only for observations *in* fold  $k$ . Then pool.

$$\hat{\theta}_{\text{DML}} = \frac{\frac{1}{n} \sum_k \sum_{i \in \mathcal{I}_k} (D_i - \hat{m}^{(-k)}(W_i)) (Y_i - \hat{\ell}^{(-k)}(W_i))}{\frac{1}{n} \sum_k \sum_{i \in \mathcal{I}_k} (D_i - \hat{m}^{(-k)}(W_i))^2}$$

This is the same device as cross-validation in Week 12, but used to protect *inference*, not to select a tuning parameter.

## The DML Recipe ★

1. Split the sample into  $K$  folds (typically  $K = 5$ ; split on the *independent unit* if data are clustered).
2. On the training folds, estimate  $\hat{\ell}(W) = \hat{E}[Y | W]$  and  $\hat{m}(W) = \hat{E}[D | W]$  with any ML method, lasso, random forest, boosting, or an ensemble.
3. On the held-out fold, form residuals  $\tilde{Y}_i$  and  $\tilde{D}_i$ .
4. Pool across folds and regress  $\tilde{Y}$  on  $\tilde{D}$ .
5. With independent observations, the standard error is the usual heteroskedasticity-robust one from that final bivariate regression: **orthogonality is what makes it valid** despite the ML first stage. *With clustered data, keeping clusters intact within folds is not enough*, the score variance must also be clustered at the independent unit.

## The DML Recipe ★ (cont.)

6. Repeat the whole split several times with different seeds and take the median; the answer should not depend on the partition.

Chernozhukov et al. (2018). In R: DoubleML with `mlr3` learners.

# What the Theory Requires, and What It Does Not ★

## Requires

- Conditional ignorability given  $W$  (unchanged!).
- Overlap:  $\hat{m}(W)$  bounded away from 0 and 1 for binary  $D$ .
- Nuisance convergence fast enough that the *product* of the two rates is  $o(n^{-1/2})$ ; each faster than  $n^{-1/4}$  is sufficient but not necessary. Approximate sparsity for lasso, smoothness for forests.
- Orthogonal moment + cross-fitting.

## Does not require

- Correct functional form for  $g$  or  $m$ .
- Knowing which of the  $p$  covariates matter.
- $p \ll n$ . High-dimensional  $W$  is the point.
- Consistent *model selection*, lasso may pick the wrong variables and  $\hat{\theta}$  is still fine.

## What the Theory Requires, and What It Does Not ★ (cont.)

The assumption that does not go away

**DML does not identify anything.** It is an estimation technology for a parameter already identified by the arguments of Weeks 5–11. If  $W$  omits a confounder, no amount of machine learning helps.

## Where DML Plugs In

The same two ingredients, an orthogonal score plus cross-fitting, apply to almost every design in this course:

| Design                             | Nuisance functions                      | Week |
|------------------------------------|-----------------------------------------|------|
| Partially linear regression        | $E[Y   W], E[D   W]$                    | 1    |
| Interactive / ATE under CIA (AIPW) | $E[Y   D, W], \Pr(D = 1   W)$           | 6    |
| Instrumental variables             | $E[Y   W], E[D   W], E[Z   W]$          | 7    |
| Regression discontinuity           | $E[Y   W]$ within the bandwidth         | 9    |
| Conditional DID                    | outcome trend and propensity, by cohort | 10   |

- For IV: residualize  $Y$ ,  $D$ , and  $Z$  on  $W$  using ML, then run 2SLS on the residuals. This is what lets us replace AJR's "linear in distance from the equator plus continent dummies" with something flexible.



## Where DML Plugs In (cont.)

- For conditional DID: this is precisely the doubly robust Callaway–Sant’Anna estimator from Week 10, with ML nuisances.

# HTE

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## From ATE to CATE ★

Drop the constant-effect assumption. The target becomes the **conditional average treatment effect**

$$\tau(x) = E[Y_i(1) - Y_i(0) \mid X_i = x].$$

- Sociologically this is often the substantive question: *for whom* does college pay off, *which* neighbourhoods benefit, *who* is harmed.
- The naive approach, run subgroup regressions, report the significant ones, is a **multiple-testing disaster** and does not replicate. It is the specification search of Week 2 with more rooms to search.
- The modern approach: build an orthogonal, doubly robust *pseudo-outcome* whose conditional mean is  $\tau(x)$ , then use ML to estimate that conditional mean, with honest sample splitting so that the subgroups are not chosen using the same data that tests them.

# The Doubly Robust Score as a Pseudo-Outcome ★

Take the AIPW summand from Week 6, unit by unit:

$$\Gamma_i = \hat{m}_1(X_i) - \hat{m}_0(X_i) + \frac{D_i(Y_i - \hat{m}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - D_i)(Y_i - \hat{m}_0(X_i))}{1 - \hat{e}(X_i)}.$$

- Its *average* is the AIPW estimate of the ATE; that was Week 6.
- With the *true* nuisance functions, its conditional mean is exactly the CATE:  $E[\Gamma_i \mid X_i = x] = \tau(x)$ . With cross-fitted estimates it holds approximately, at a rate governed by the same product condition as before.
- So: construct  $\Gamma_i$  with cross-fitted nuisances, then **regress  $\Gamma_i$  on  $X_i$  by any method you like**.
  - Regress on a few chosen covariates → **group average treatment effects** (GATEs) with valid standard errors.
  - Regress with a forest → a nonparametric CATE surface.
  - Regress on a constant → back to the ATE.
- **This one construction unifies the whole section.**

## Causal Forests ★

Wager and Athey (2018), Athey, Tibshirani, and Wager (2019): a random forest whose splits are chosen to maximize *heterogeneity in the treatment effect* rather than in the outcome.

- **Honesty:** within each tree, one subsample chooses the splits and a *disjoint* subsample estimates the effects in the resulting leaves. This is what delivers asymptotic normality and valid pointwise intervals, without it, leaves are selected *because* they show large effects.
- **Local centering:** residualize  $Y$  and  $D$  on  $X$  first (i.e. orthogonalize), exactly as in DML.
- **Output:** an estimate  $\hat{\tau}(x)$  for every unit, with standard errors, plus variable-importance measures for *effect* heterogeneity.
- `grf::causal_forest()`.

# Reporting Heterogeneity Honestly ★

An estimated  $\hat{\tau}(x)$  that varies is not by itself evidence of heterogeneity, noise varies too.

- **Calibration test:** regress the doubly robust scores on  $\hat{\tau}(X_i)$  (out of fold). A slope near 1 indicates real, well-calibrated heterogeneity; a slope near 0 says your CATE surface is noise. `grf::test_calibration()`.
- **Sorted group ATEs:** rank units by  $\hat{\tau}$ , form quantile groups, and report the doubly robust ATE *within each group* using held-out data. Then test whether the top and bottom groups differ.
- **Pre-specify** the moderators you care about wherever possible, and label exploratory heterogeneity as exploratory.
- **Effect heterogeneity is where causal ML is most useful and most abused.** The honest-splitting discipline is not optional.

## Where This Leaves the Course

- **One idea, three appearances.** FWL (Week 1)  $\rightarrow$  double robustness and Neyman orthogonality (Week 6)  $\rightarrow$  DML and causal forests (today). Each generalizes the last; none replaces identification.
- **ML relaxes functional-form assumptions, not identifying assumptions.** That is a real and substantial gain, many published identification arguments lean on linearity they never defended, but it is a bounded one.
- **The binding constraint is still the design.** A causal forest on a confounded comparison estimates heterogeneity in a bias.

### Next week

Synthesis, and a review for the final exam.

# References

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